How Often Must the Cap Return?

A fixed-cap timing frontier after a one-time handover loses protection

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1 Hold the cap fixed; change its timing

Two shops choose prices simultaneously. Between cap rounds, a guard allows any price whose modeled immediate profit against the previous rival price is within epsilon of the best profit. A real cap execution instead permits the full price interval up to the original physical cap, independently of old prices. It replaces the guard on that execution; it is not their intersection. A cap is applied immediately and then every K executions, with K-1 ordinary guard executions between caps.

The preceding transfer study found that a finite initialization cannot protect a target above the unchanged guard's continuous worst long-run floor. We now test an ongoing intervention: the SAME inherited physical cap. Eight original labels, five unchanged targets and two tolerance semantics are fixed before the run (80 comparisons). The 90/95/99/100% labels refer to each original permanent-cap surplus floor; the anchor label uses the certified lower bound of the old near-Nash anchor. Requested values are relaxed by their saved allowance 1e-8*max(1,requested value); all comparisons retain the resulting saved thresholds. These labels include equivalent calibrations; they are not independent economic replications.

Base example. For the original 95% target minus margin, F approximately 68.513088, a cap every 4 executions guarantees approximately 68.675931 customer surplus on average. Every 5 is insufficient: its attained worst average is approximately 68.228977. This is maximality within the fixed-cap, fixed-period family, not over all controllers.

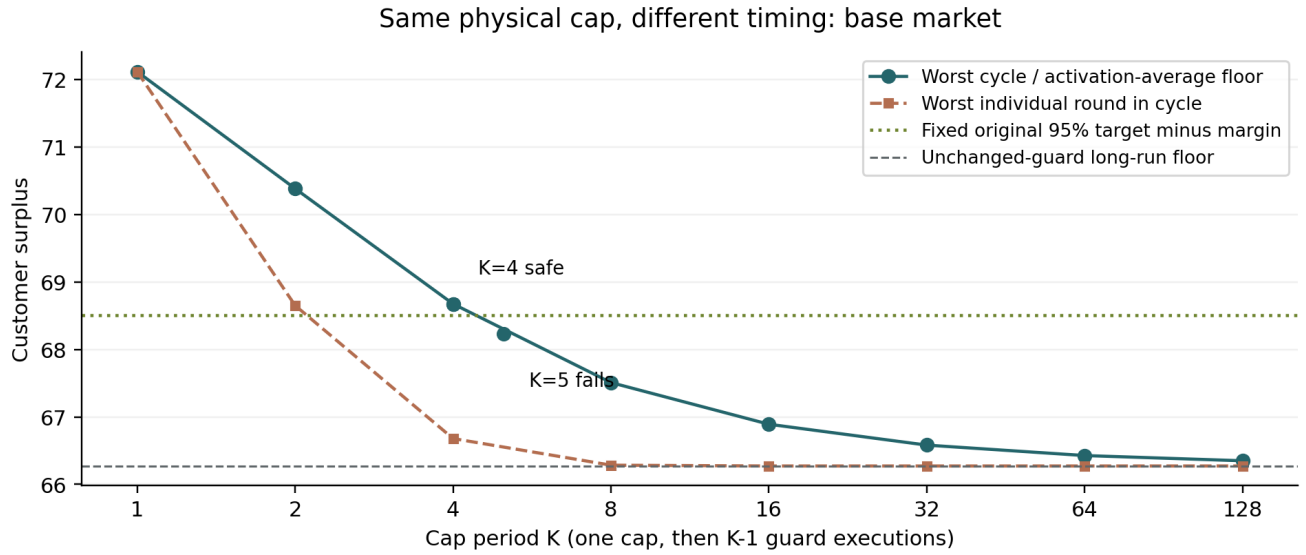


Figure1. Validated effective-continuum values, shown approximately. The cap ceiling and economic parameters stay fixed. A cycle-average guarantee differs from its worst individual round; the latter can fall below the protected average.

The negative result matters. The 99% and 100% target labels require a cap every execution in all eight labels. Seven 90% targets already hold without recurring intervention; the doubled-tolerance label requires K=17. Recurrence is not universally light-touch.

2 Why the worst cycle is exact

Write normalized prices x=(p-c)/mu. The inherited continuous analysis validates a nondecreasing upper permitted-price map U with contraction factor rho<1 and unique fixed point q_star. Customer surplus C falls in either price. Normalize the inherited physical cap as A=(p_cap-c)/mu and verify ell<=A<q_star. Then a_0=A, a_(j+1)=U(a_j) increases to q_star, and r_j=C(a_j,a_j) decreases to C_star.

On a cap round, every permitted pair is at most(A,A). By induction and monotonicity, every permitted pair at phase j is at most(a_j,a_j). Thus its customer surplus is at least r_j; selecting both upper allowed prices attains equality. This proves an all-permitted-policy guarantee and an attaining adverse path. It does not assert that independent sellers choose that path.

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G_K = (r_0 + ... + r_(K-1))/K
Worst cumulative CS at T=m*K+s:
m*(r_0 + ... + r_(K-1)) + (r_0 + ... + r_(s-1))
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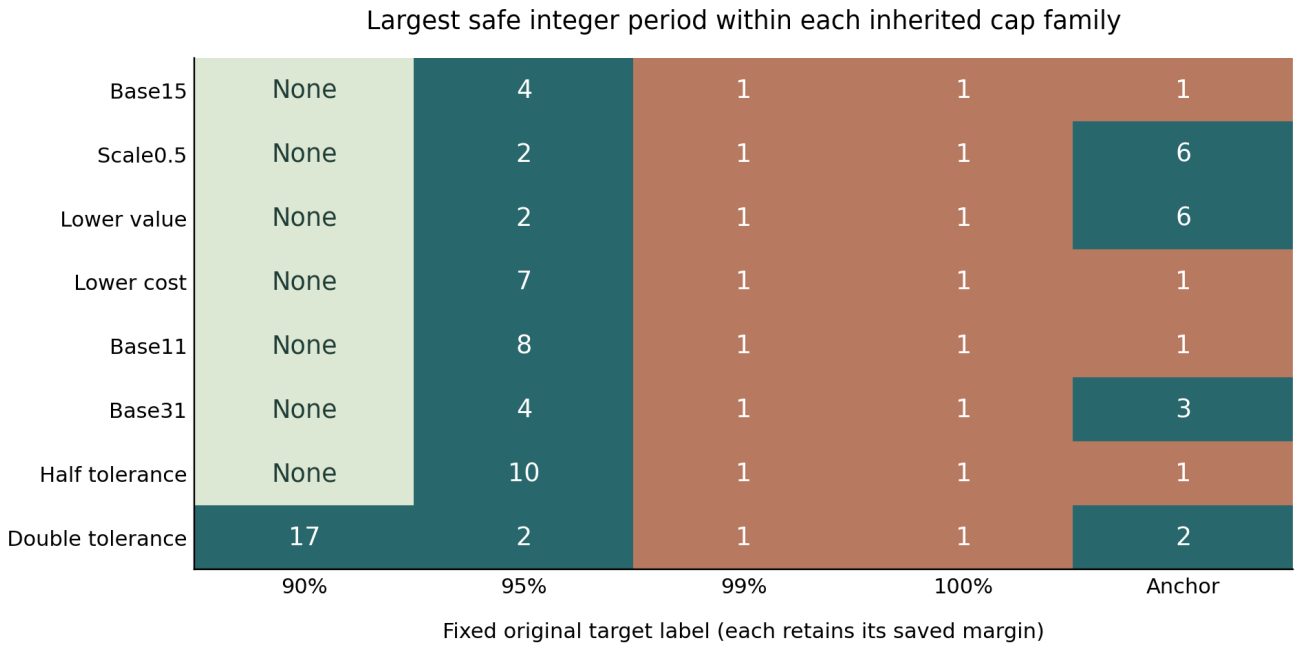
Because the rewards descend, EVERY finite average measured from the first actual cap execution is at least G_K, as is the long-run average. An old price pair earns no reward. Observation starting at an arbitrary phase has only the long-run guarantee unless the initial partial cycle is separately bounded. A single round can fall below G_K.



Figure2. The attained worst path for the base effective K=4 controller. Green bands mark actual cap executions. Cumulative averages begin with the rewarded cap, not an artificial initial-state observation.

G_K strictly decreases to C_star. The largest safe integer requires both G_K.lower>=F and G_(K+1).upper<F. We compute 64 outward-bounded orbit terms using separate signed endpoint solves, then use a contraction/Lipschitz geometric tail. The declared search ceiling is 65536; a safe ceiling would remain censored rather than imply an evaluated 65537. No case was censored or unresolved here. The nominal epsilon and primary effective exact-real epsilon+tau (imported tau=1e-12) yield the same integer classifications. These continuous semantics do not exactly replay every binary64 profit operation.

3 Keep every target, including unhelpful outcomes



None: no recurring cap for long-run goal. 1: cap every execution.
Same nominal/effective classifications; equivalent labels are not independent economies.

Figure3. Every inherited scenario/target label is retained. Entries are largest certified periods, not learned choices. None means the unchanged guard already meets the long-run target: zero startup caps suffice for that objective, whereas one immediate cap additionally protects all finite activation averages. Physical cap prices can differ across original grid-size labels; cross-label timing differences cannot be attributed solely to resolution.

4 A separately certified, playable numerical game

The graphical game uses 61 base quotes and prioritizes the EXACT inherited cap over any base quote within the predeclared 1e-12 absolute distance, then sorts and deduplicates (61/62 quotes). Its profit comparisons and allowed masks follow the specified binary64 calculation. Saved customer-surplus weights are then treated as exact rational numbers. Full-table monotonicity and the upper-map conditions are checked separately. Its certificate is for that numerical finite game, not an exact-real continuous logit certificate.

For a general finite cycle the all-activation-average floor is H_K=min_(1<=s<=K)(r_0+...+r_(s-1))/s. This also handles cycles that do not descend. The saved cases independently verify descending rewards, so H_K=G_K here. The game exposes both shops' permitted prices, expected purchases, individual profits, cap clock, actual executions, undo/reset and separate manual/scripted ledgers. The scripted maximum-price benchmark uses no model player and is not a profit-maximizing strategy.

Independent checks. Continuous audit: 16 models, 80 target labels, 2,048 scalar root certificates and 44,110 checks; five corruption controls rejected. Finite audit: 30,506 payoff pairs, 191,823 checks and 4,852 actual executions; four controls rejected. Independent Fraction code verifies all 64 finite cycle/prefix certificates and rejects three corruptions. Controller logic and 130-digit interval code are separate from the primary 80-digit implementation; both interval stacks share Python Decimal transcendental arithmetic. This is not formal verification or independent arithmetic-library validation.

5 Contribution, limits and reproducibility

Periodic resetting, intermittent control, frame-average optimization and ongoing platform intervention are established. Our narrow contribution is a tight all-permitted-policy timing frontier for the SAME inherited cap in this lagged tolerance-guard model, paired with its finite-activation accounting and a separately certified playable game. The mechanism is a model-specific synthesis; the targeted literature review establishes neither first publication nor general controller optimality.

Relevant primary sources: De Bruyne and Mori(2023), Physical Review Research 5,013122,doi:10.1103/PhysRevResearch.5.013122 (reset optimization); Liang and Xia(2020), Advances in Difference Equations 576,doi:10.1186/s13662-020-02969-3 (periodic intermittent control); Neely(2021), JMLR 22(279):1-44 (renewal reward/time objectives); Brero et al.(2022), NeurIPS 35, "Learning to Mitigate AI Collusion on Economic Platforms" (ongoing platform commitments). MacKay and Weinstein(2022), Washington University Law Review 100,111, "Dynamic Pricing Algorithms, Consumer Harm, and Regulatory Response," discusses algorithmic price regulation and timing. The separate close learned-policy audit by Lopes de Moraes and De Losso(2026), arXiv:2610.21867v2, prevents novelty claims about unilateral-DP/basin audits. Detailed source links and inspected sections are retained in the novelty notes.

All economic parameters, enforcement, symmetry and phase timing are known. No uncertainty in real demand or guard enforcement is included. A maximum-price stress path can be feasible without being stable against a seller's strategic deviation. No autonomous collusion discovery, incentive-compatible equilibrium, human comprehension, real-market welfare effect or adaptive-controller optimum is established. Cap-rule rate 1/K counts rule applications; it is not the fraction of quotes changed or a measured economic intervention cost. Profits on the worst-CS path are descriptive, not uniform seller-profit guarantees.

The practical next checks are model-misspecification margins and whether the adverse script is attractive to individually optimizing sellers. Neither is answered by drawing another permitted path. The finite/continuous distinction, one-time/recurring distinction and individual-round/activation-average distinction remain explicit throughout the evidence.

Reproduction and disclosure

Protocol/theory, source and old-input hashes were frozen before each numerical run. Exact imported binary64 constants, roots, directed intervals, all 80 comparisons, period boundaries, per-case journals and resource records are saved. Read-only independent checks: python research/audit-sparse-reset.py; node research/audit-sparse-display.js; python research/audit-sparse-reset.py --finite. Rebuild the displayed figures with python research/build-recurring-figures.py and this note with python research/build-recurring-note.py. Earlier closed reports and artifacts are preserved.

Codex agents substantially assisted formulation, code, independent AI-agent auditing, interpretation and writing under Mohammadpayam Teabi's direction. Agent agreement is not human peer review. This stage used no model/account-backed game players, paid APIs, external customer data, public deployment or submission. Manual and automated interface checks do not constitute a human study.